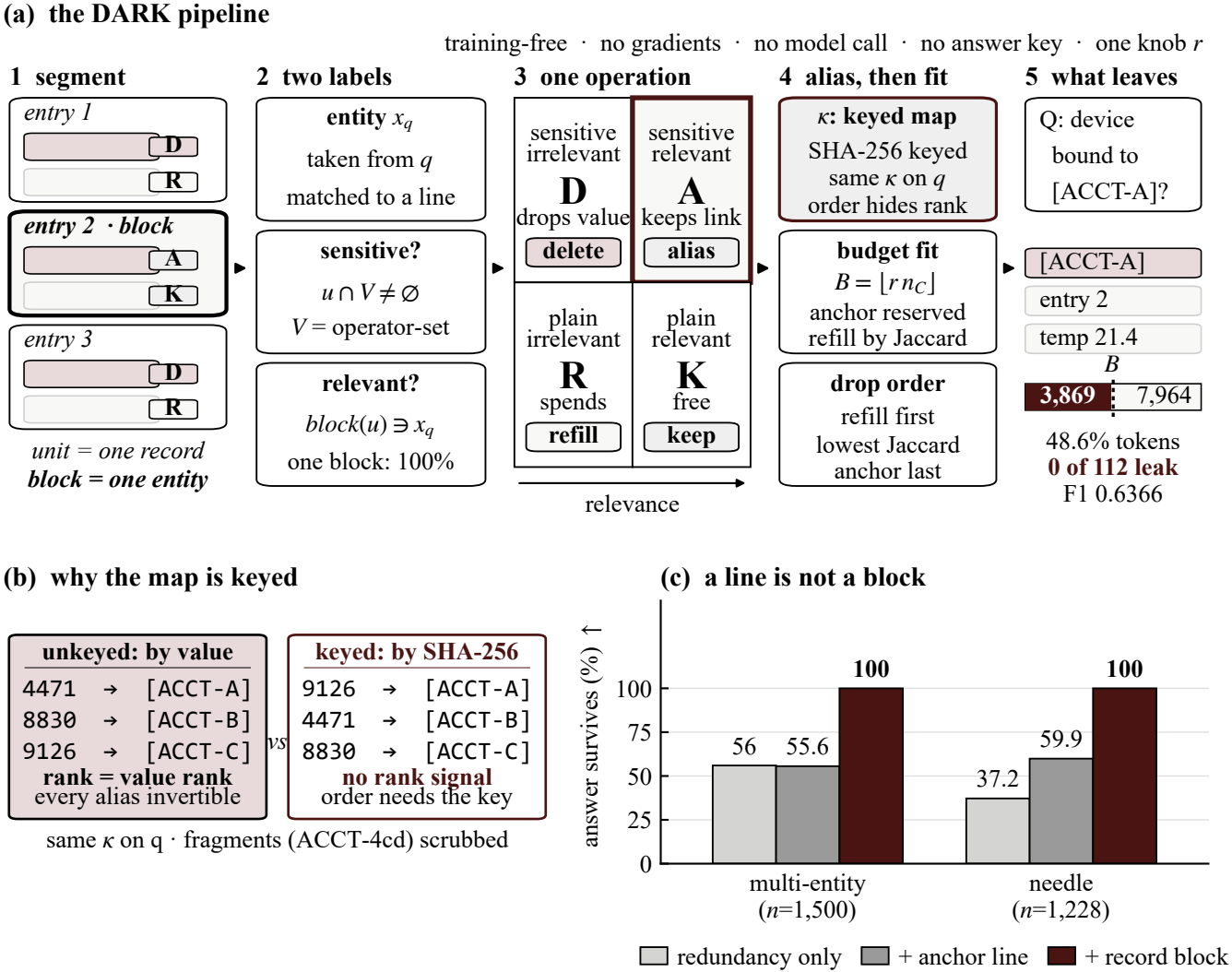

(a) the DARK pipeline
training-free · no gradients · no model call · no answer key · one knob r
1 segment
entry 1
D
R
entry 2 · block
A
K
entry 3
D
R
unit = one record
block = one entity
2 two labels
entity x_q
taken from q
matched to a line
sensitive?
$u \cap V \neq \varnothing$
V = operator-set
relevant?
$block(u) \ni x_q$
one block: 100%
3 one operation
sensitive irrelevant
D
drops value
delete
sensitive relevant
A
keeps link
alias
plain irrelevant
R
spends
refill
plain relevant
K
free
keep
relevance
4 alias, then fit
κ: keyed map
SHA-256 keyed
same κ on q
order hides rank
budget fit
$B = \lfloor r\, n_C \rfloor$
anchor reserved
refill by Jaccard
drop order
refill first
lowest Jaccard
anchor last
5 what leaves
Q: device bound to [ACCT-A]?
[ACCT-A]
entry 2
temp 21.4
B
3,869
7,964
48.6% tokens
0 of 112 leak
F1 0.6366
(b) why the map is keyed
unkeyed: by value
4471 → [ACCT-A]
8830 → [ACCT-B]
9126 → [ACCT-C]
rank = value rank
every alias invertible
vs
keyed: by SHA-256
9126 → [ACCT-A]
4471 → [ACCT-B]
8830 → [ACCT-C]
no rank signal
order needs the key
same κ on q · fragments (ACCT-4cd) scrubbed
(c) a line is not a block
answer survives (%) →
0
25
50
75
100
56
55.6
100
37.2
59.9
100
multi-entity
(n=1,500)
needle
(n=1,228)
redundancy only
+ anchor line
+ record block